



Module Title

Cybersecurity and Networking

Internship Assignment Submission

Security Misconfiguration & Hardening

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1. ASSIGNMENT A02:2025 SECURITY

MISCONFIGURATION

1.1 Answer all questions in your own words.

1.1.1 PART A — FUNDAMENTALS

1.What is Security Misconfiguration?

Security Misconfiguration means setting up a system, application, or server in an unsafe way that can create security risks.

2. Give three examples of security misconfiguration.

- Keeping default accounts or passwords.
- Leaving Debug Mode enabled in production.
- Running unnecessary services or features.

3. Why can default accounts be dangerous?

Default account can be dangerous because their usernames and passwords may be known. Attackers can try these details to gain access.

4. Why should unnecessary features and services be disabled?

Unnecessary features and services can create extra security risks. Disabling them reduces the attack surface of the system.

5. What is application hardening?

Application hardening means making an application more secure by removing unnecessary features, restricting access, and using secure settings.

6. What is a configuration baseline?

A configuration baseline is a standard secure setup used as a reference. It helps organization check whether systems are configured correctly.

7. What is configuration drift?

Configuration drift happens when a system's settings change over time and become different from the approved secure configuration.

8. Why should production and development configurations be separated?

Development environments may use test accounts, debug mode, and detailed errors. These settings may create security problems if they are used in production.

9. Why can verbose error messages create security risk?

Detailed error message can reveal information such as file paths, software versions, or database details. Attackers may use this information to understand the system.

10. Explain the Principle of Least Privilege in configuration.

Least Privilege means giving a user, application, or service only the permissions needed to do its job.

1.1.2 PART B — WEB CONFIGURATION

11. What is Debug Mode?

Debug Mode is a feature that helps developers find and fix problems by showing detailed error information. It should normally be disabled in production.

12. Why should debugging information not normally be exposed publicly in production?

It can reveal sensitive technical information about the application or server. Attackers could use this information to find possible weakness.

13. What are HTTP security headers?

HTTP security headers are instructions sent by a website to a browser to provide additional security protections. Examples include HSTS and X-Content-Type-Options.

14. What is the basic purpose of HSTS?

HSTS tells compatible browsers to use HTTPS when connecting to the website according to its security policy.

15. What is the basic purpose of X-Content-Type-Options: no sniff?

It tells the browser not to guess the content type in certain situations. This helps prevent some unsafe content-handling behavior.

16. Does using security headers alone guarantee that a website is secure? Explain.

No. Security headers are only one part of web security. Other areas such as authentication, authorization, secure-coding, and server configuration are also important.

1.1.3 PART C — CLOUD & SECRETS

17. Give one example of a cloud configuration mistake.

One example is making private cloud storage publicly accessible by mistake.

18. Why should API keys and passwords not be publicly exposed?

Attackers could use exposed API keys or passwords to access systems, accounts, or sensitive information.

19. Are environment variables automatically secure simply because they are environment variables?

No. Environment variables can still contain sensitive information. They must be properly protected and access to them should be restricted.

Explain. 20. Why should cloud permissions follow Least Privilege?

Cloud users and services should only have the permissions they need. This limits the damage if an account or service is compromised.

1.1.4 PART D — CONFIGURATION REVIEW

1. Identify every unsafe configuration

The unsafe settings are:

- Debug Mode is ON.
- Default Admin Account is enabled.
- Detailed errors are visible to the public.
- Unused Demo App is installed.
- Unused Service is running.

2. Identify the settings that are not automatically a problem

- HTTPS: ON – This is normally a good security setting.
- Logging: Enabled – Logging is normally useful for monitoring and investigating problems.

3. Give at least four hardening recommendations

- Turn off Debug Mode
- Disable or remove the default admin account.
- Hide detailed error from public users.
- Remove the unused demo application
- Disable the unused service.

4. Which OWASP Top 10:2025 category applies?

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5. Explain Why.

The system has several unsafe settings, such as Debug Mode, default accounts, detailed errors, and unused services. These settings can increase the attack surface and create security risks.

1.1.5 PART E — MINI CASE

A fictional company stores internal documents in cloud storage. The storage configuration accidentally allows public Internet access.

Answer:

1.What went wrong?

The company's cloud storage was incorrectly configured to allow public Internet access.

2. Which OWASP category applies?

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3. What asset is at risk?

The Company's internal documents and confidential business information are at risk.

4. What could the impact be?

Unauthorized people could view or download the company's internal documents. This could cause information leakage and other security problems.

5. Give a high-level defensive recommendation.

The storage should be changed to private access, and only authorized users should be given the required permissions. Cloud access should also be reviewed regularly.